Beyond Boundaries — verdict

Do the Xenium Multimodal Cell Segmentation images add anything beyond segmentation? Mouse EAE spinal cord, RRMAP2 runs 1–6: 54 images, 67 animals, 1.38 M cells. Status 8 October 2026; details in `BeyondBoundaries_methods_results.pdf` (methods + all numbers) and `BeyondBoundaries_lesion_story.pdf` (biology).

> **A modest yes.** The images are first of all a segmentation aid. Beyond that they give a small number of robust readouts the 5K panel cannot. Their pooled channels and optical bleed limit them as a protein assay, so specific protein questions need dedicated stains.

What holds (replicated across runs, with controls)

Readout	Evidence	Use
Segmentation (the kit's purpose)	18S interior stain draws ~95 % of cell masks	real; not benchmarked against alternatives here
Neuropil loss at the lesion's grey/white-matter border (ATP1A1 around each cell; *Atp1a1* not on the panel)	~15–20 % less ATP1A1 in lesions within ~75 µm of grey matter than in healthy WM at the same distance, both run sets (×0.78, 25/31 animals; ×0.83, 13/18); **none deeper in white matter**	damage the panel can't show; neuron-derived RNA between cells is lower at the same place (nb 23). **Corrected 8 Oct:** the earlier "~17 % WM loss" was mostly the ATP1A1 gradient away from grey matter (nb 20 §9)
Astrocyte reactive RNA without vimentin protein in active disease, consistent with RNA preceding protein (stages from different animals) (*Vim* not on the panel)	protein vs RNA ρ 0.46–0.58 per run; "RNA-only" astrocytes enriched in active disease in both run sets	the clearest protein-vs-RNA insight
Images alone map lesions and cell types on unseen runs	lesion AUROC 0.85–0.91, cell type 0.42–0.47 (chance 0.07), leave-one-run-out over 5 runs	QC and annotation cross-check; redundant with RNA for biology

Human MS spinal cord (one run, 4 sections): CD45 **is** detectable on lymphocytes there (boundary-channel-specific at matched size, tracks *PTPRC*, outlines perivascular cuffs), and the αSMA/vimentin channel follows *VIM* RNA (ρ 0.69–0.98), validating the vimentin readout. Neuropil loss is inconclusive in human.

What is weak or partial

- **Lesion vimentin and outcome:** low at peak, fills lesion interiors, tends to be higher in animals that do well (ρ ≈ -0.45 after time adjustment). But part of it is shared with VSMCs in the same channel (astrocyte-specific ρ ≈ -0.33), it is entangled with disease burden, and the same-image test is weak (12/19 pairs). **Weak to moderate.**
- **18S texture as a severity marker:** replicates in astrocytes, myeloid cells and oligodendrocytes only.
- **18S gain in activated lesion cells beyond RNA content:** plausible, not stress-tested.

What does not work, or was withdrawn after controls

- **CD45** is undetectable in mouse spinal cord (the boundary channel reports ATP1A1 only); in human it works.
- **No hidden cell states:** ~85 % of image variance isn't explained by RNA, but it is mostly a local optical/staining field.
- **Images don't improve lesion identity or severity prediction** over the transcriptome.
- **Withdrawn:** nuclear RNA retention in lesion oligodendrocytes (white/grey-matter mix; notebook 23); T cells orienting 18S towards vessels (bleed from bright neighbours + outline geometry); a vimentin ring at lesion edges and vimentin-border containment (artefacts of a lesion-object definition). No evidence that vimentin is a barrier to immune cells (within-lesion test ρ +0.035, 95 % CI -0.016 to +0.073).

Why: three structural limits

1. **Pooled channels** (αSMA + vimentin; ATP1A1 + CD45 + E-cadherin): every protein claim carries "or the other one".
2. **Optical bleed and outline geometry:** small cells next to bright neighbours borrow their signal; polarity and "around the cell" measures need brightness-matched and nucleus-relative controls.
3. **The stains were chosen to draw cell boundaries**, not to answer biological questions.

Recommendations

- **Keep the kit** for segmentation; use border neuropil loss and astrocyte vimentin state as add-on readouts, with the caveats above.
- **For protein biology:** post-Xenium immunofluorescence on the same sections with single-target antibodies (vimentin, αSMA, GFAP; a mouse-reactive CD45). The most informative sections: MILD16 vs SEVERE16 (they share images) and MONOPHASIC vs REMISSION1.
- **Future panels:** add *Vim*, *Acta2*, *Gfap*, *Atp1a1* as custom genes.

The bigger contribution may be the disease biology (transcriptome-led, all five runs)

Lesions defined against controls (controls ~1 %, not 4–20 %) resolve along a clear course: active monocyte-derived infiltrate → fibrotic, demyelinated and glial-reactive tissue. Chronic severity is set at the first attack and means lesions that stayed active. Relapse can't be predicted clinically; B cells accumulate after the first attack in meningeal aggregates (fewer in never-relapsing animals), and relapses are mostly new lesions.

For discussion: the paper's framing; slides for post-Xenium immunofluorescence; which open threads to close first.